Capacity and Service-Level Stability in Quick Commerce Delivery Networks: A System Dynamics Study

Team Name: System Thinker

Team Members:
Ashish Shukla (25M1542)
Bhupesh Ranjan Jha (25M1541)
Motupalli Hemanth Kumar (25M1512)

Scenario 1: Demand Surge

To analyze system behavior under increased demand, the base demand was increased from 25 to 40 orders per hour.

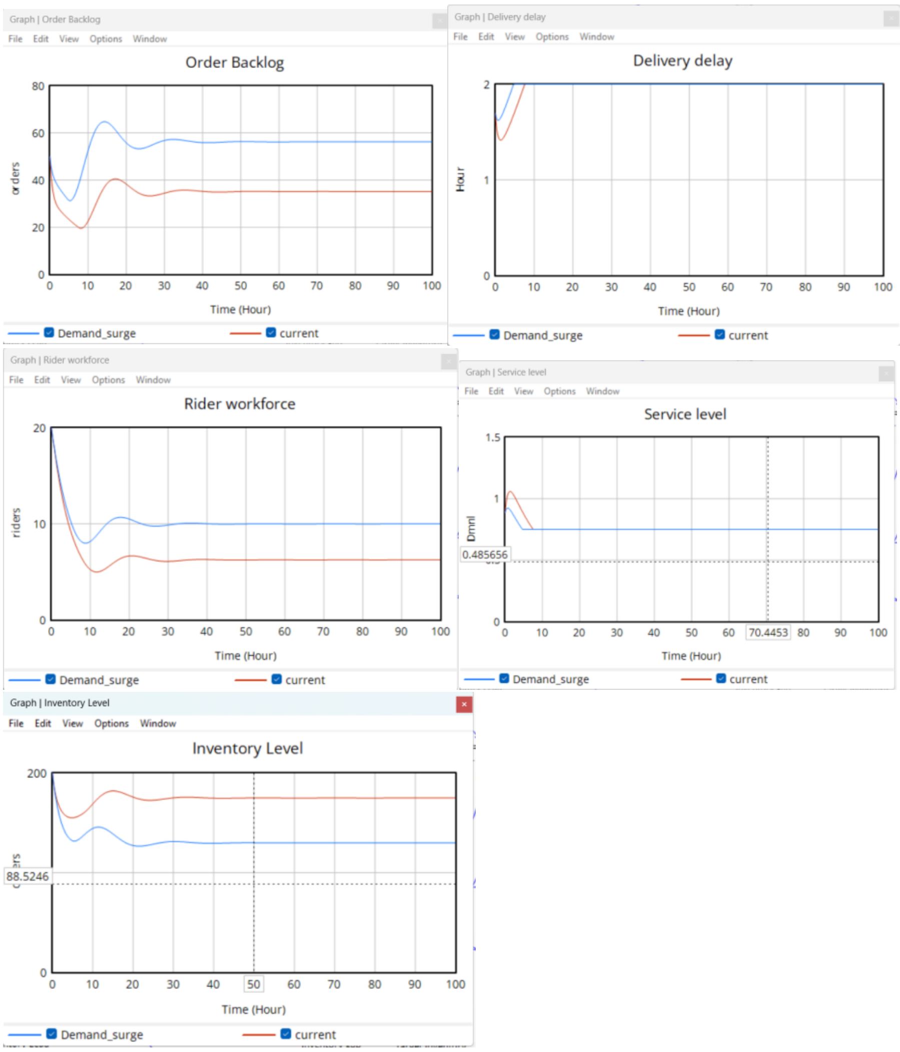
The results show that the order backlog increases significantly, exhibiting higher peaks and larger oscillations compared to the baseline scenario. The system stabilizes at a higher backlog level, indicating that fulfillment capacity is insufficient to match the increased demand.

Delivery delay rises sharply and quickly reaches its maximum level due to increased capacity utilization. As a result, the service level declines and stabilizes at a lower value, reflecting deterioration in customer experience.

The rider workforce increases in response to higher backlog levels; however, due to delays in workforce adjustment, the system is unable to respond immediately. This leads to temporary imbalances and oscillatory behavior.

Inventory levels decline more sharply under increased demand and stabilize at a lower level, indicating increased pressure on the replenishment system.

Overall, the demand surge scenario demonstrates that the system becomes more congested and operates at a degraded performance level. The interaction of delayed workforce adjustment and inventory replenishment leads to persistent backlog and reduced service quality.



Real-World Relevance:

A similar situation is commonly observed during **festive sales or sudden demand spikes** in quick commerce platforms such as Blinkit, Zepto, and Instamart. For instance, during events like **Diwali or heavy rainfall days**, order volumes surge unexpectedly.

During such periods:

- Delivery times increase significantly
- Riders become overloaded
- Customers experience delays and cancellations

Companies often struggle to scale operations instantly due to hiring and onboarding delays, leading to backlog accumulation and degraded service levels.

Scenario 2: Peak Demand with Recovery

To capture more realistic demand fluctuations, a time-varying demand pattern was introduced where demand increases in stages and subsequently decreases.

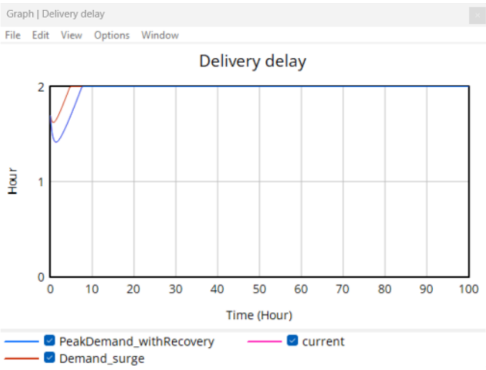
The results show that the system experiences multiple cycles of backlog accumulation and clearance. Each increase in demand generates a surge in backlog and delivery delay, while the subsequent decline in demand allows the system to recover, reducing backlog levels.

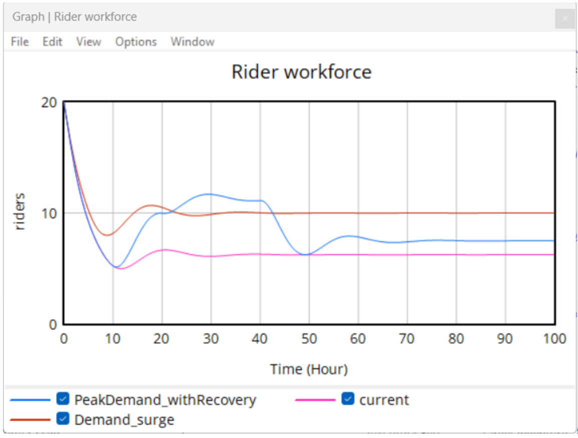
Compared to the demand surge scenario, the system does not settle at a permanently high backlog level. Instead, it demonstrates recovery behavior, indicating partial resilience to fluctuating demand.

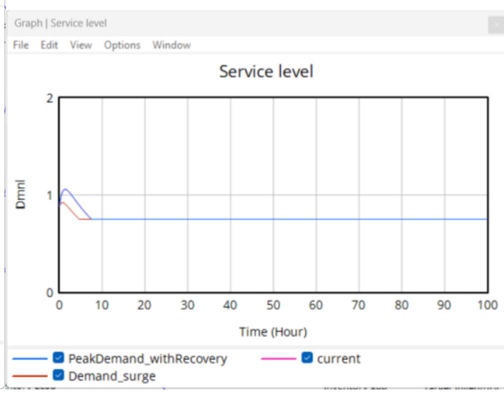
However, the rider workforce exhibits overshooting behavior. Due to delays in workforce adjustment, hiring continues even after demand decreases, resulting in temporary overcapacity before stabilizing.

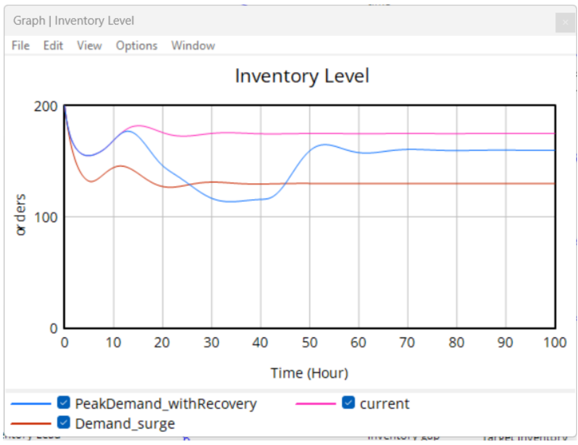
Inventory levels also show cyclical behavior, declining during peak demand periods and recovering as replenishment processes respond to reduced demand.

Overall, this scenario highlights the dynamic nature of the system and reveals both its ability to recover from demand shocks and its limitations due to delayed adjustment mechanisms.









Real-World Relevance:

This scenario reflects typical daily demand patterns observed in quick commerce platforms, where order volumes fluctuate throughout the day, with significant peaks during evening hours and lower demand during off-peak periods. Platforms such as Blinkit and Zepto regularly experience these cycles, leading to temporary increases in backlog and delivery delays during peak times, followed by recovery as demand subsides. However, due to delays in workforce adjustment, companies often face challenges such as overstaffing after peak periods and inefficiencies in resource utilization, which is consistent with the behavior observed in the model.

Scenario 3: Flexible Workforce Policy

To improve system responsiveness under fluctuating demand conditions, a flexible workforce policy was introduced by incorporating a surge factor that temporarily increases workforce levels during peak demand periods.

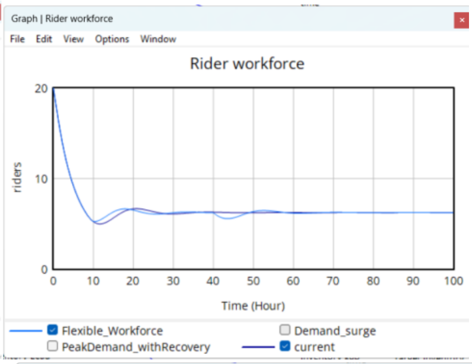
The results show a significant improvement in system performance compared to previous scenarios. The order backlog is substantially reduced, with lower peak values and faster stabilization. The system is able to maintain backlog levels close to the base case, indicating effective demand handling.

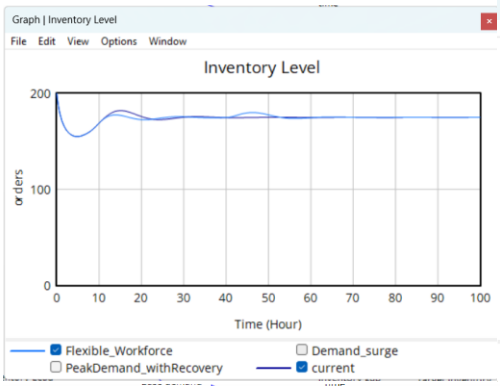
Delivery delay is also better controlled, reducing congestion during peak periods. Consequently, the service level stabilizes more quickly, ensuring improved customer experience.

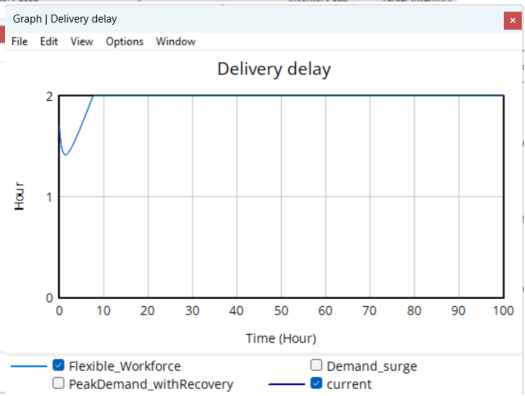
The rider workforce adjusts dynamically, increasing during peak demand and stabilizing without significant overshoot. This represents a more controlled and proactive workforce management strategy compared to the delayed and reactive adjustments observed earlier.

Inventory levels remain relatively stable, with minimal depletion during peak periods, indicating improved coordination between demand and fulfillment processes.

Overall, the flexible workforce policy enhances system stability, reduces oscillations, and improves service performance. This demonstrates the effectiveness of proactive capacity management strategies in handling demand variability in quick commerce systems.







Real-World Relevance:

This scenario represents the flexible workforce strategies widely adopted in modern quick commerce systems, where companies use gig-based delivery models to dynamically adjust capacity. Platforms like Blinkit, Zepto, and Swiggy Instamart employ surge pricing and incentive mechanisms to attract additional riders during peak demand periods, allowing them to handle increased order volumes efficiently. As demand decreases, workforce levels naturally adjust without long-term overstaffing, improving operational efficiency. This real-world approach aligns closely with the model results, where proactive workforce scaling helps reduce backlog, control delivery delays, and maintain service levels.

What if Scenario 4 : Workforce Adjustment Time

To analyze the impact of workforce responsiveness, the workforce adjustment time was reduced from 5 hours to 3 hours while keeping all other parameters constant.

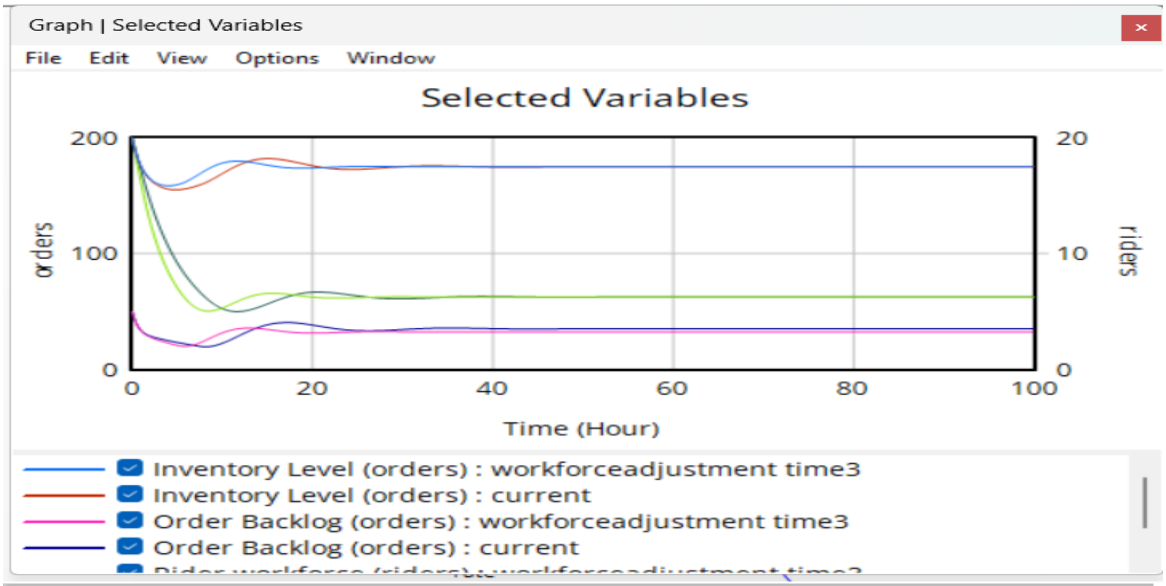
The results show that the order backlog exhibits significantly lower peaks and stabilizes more quickly compared to the base case. In the original model, delayed hiring leads to prolonged backlog accumulation and oscillatory behavior, whereas faster adjustment enables quicker response to demand fluctuations and reduces system imbalance.

Delivery delay is also reduced, as the system is able to match demand with available delivery capacity more efficiently. Consequently, service levels improve and stabilize earlier. The rider workforce increases more rapidly in response to backlog and reaches the required level without significant delay.

Inventory levels remain relatively stable in both cases; however, faster workforce adjustment reduces initial fluctuations and improves coordination between fulfillment and replenishment processes.

Overall, the scenario demonstrates that reducing delays in workforce adjustment improves system responsiveness, reduces oscillations, and enhances service performance without changing system structure.

Real-World Relevance:
This situation is commonly observed in quick commerce platforms such as Blinkit and Zepto, where faster onboarding of delivery partners or use of flexible gig workers helps companies respond quickly to demand fluctuations. Platforms that are able to scale workforce rapidly experience lower delivery delays and better service reliability compared to those with slower hiring processes.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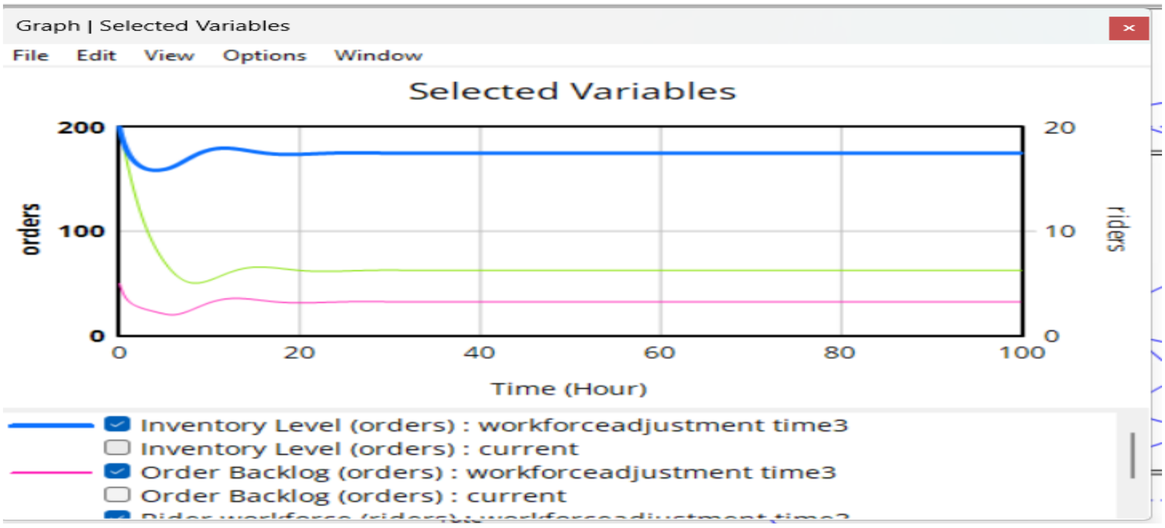
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What if Scenario 5 : Service-Sensitive Demand

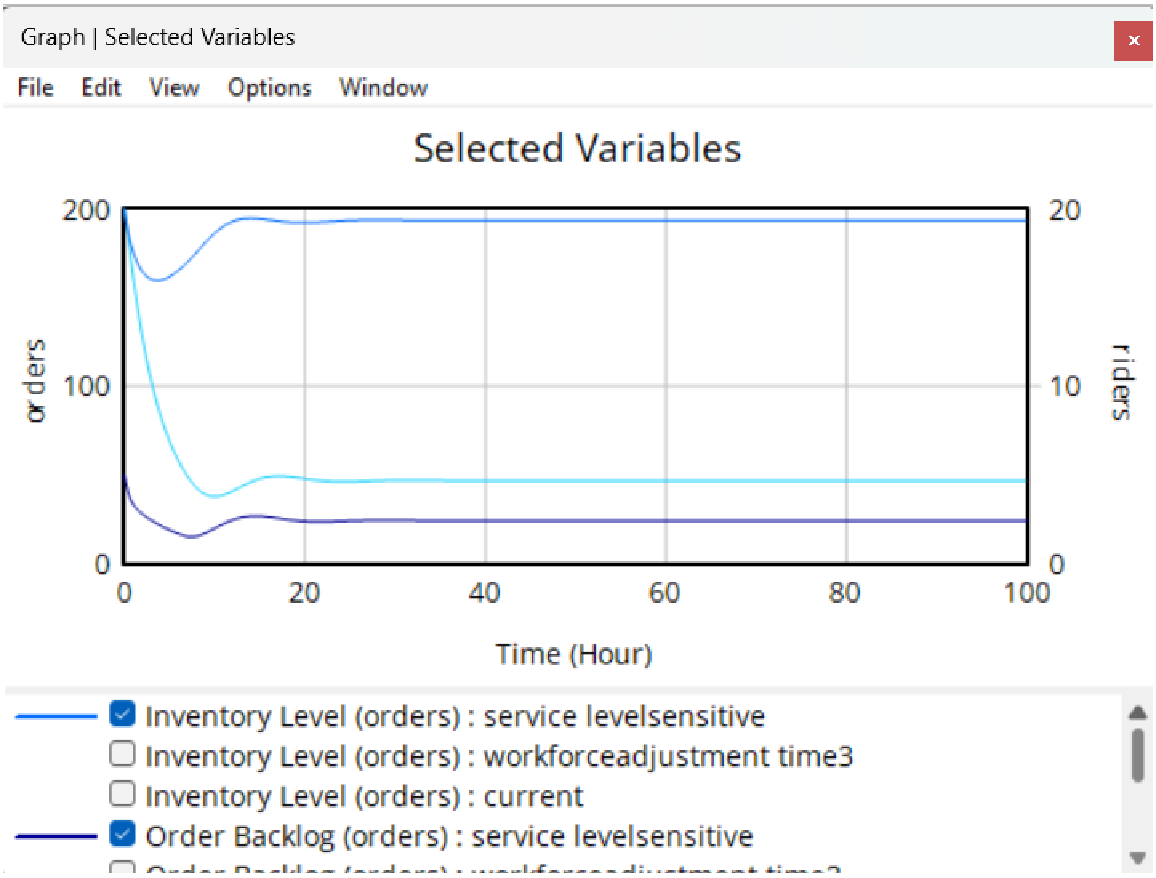
To examine the effect of customer behavior on system performance, the demand function was modified from a linear relationship with service level to a nonlinear relationship, where demand is proportional to the square of service level, while keeping all other parameters constant.

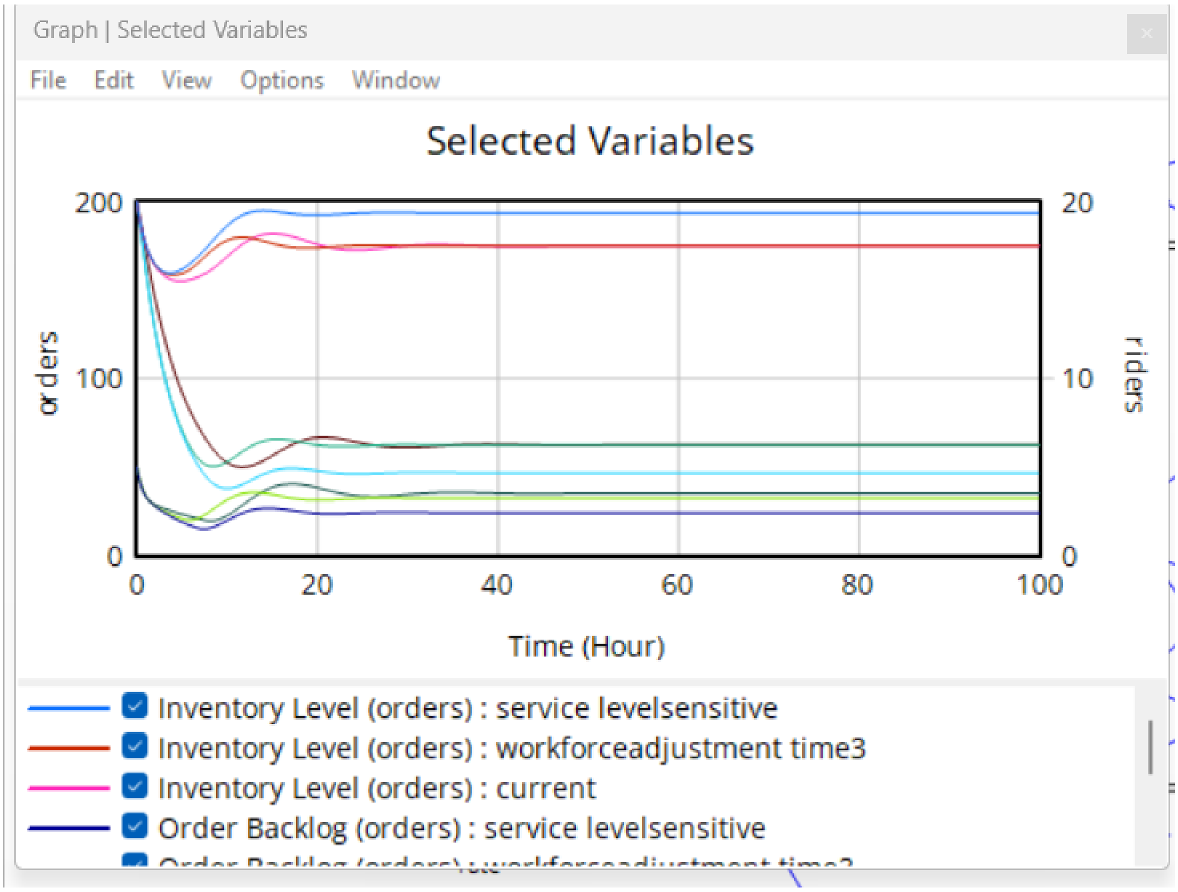
The results show that the order backlog reduces more quickly and stabilizes at a lower level compared to the base model. In the original case, demand remains relatively stable even when service deteriorates, leading to continued system pressure. In contrast, the modified model shows that demand decreases sharply when service levels drop, reducing the load on the system.

Delivery delays are better controlled, as reduced demand prevents excessive congestion. Service levels stabilize faster due to lower system stress. Inventory levels also show smoother behavior with fewer fluctuations, indicating improved balance between demand and supply.

Overall, the system exhibits faster stabilization and reduced oscillations compared to the base case, highlighting the impact of demand responsiveness on system dynamics.

Real-World Relevance:
In real-world quick commerce systems such as Swiggy Instamart and Blinkit, customers are highly sensitive to delivery delays. When service quality drops, users often reduce ordering frequency or switch platforms. This behavior reduces demand pressure on the system, similar to the stabilizing effect observed in the model.





What if Scenario 6: Rider Attrition due to High Work Pressure

To simulate the impact of workforce instability, the attrition fraction was increased from 0.05 to 0.15, representing higher rider exits due to work pressure, while keeping all other parameters constant.

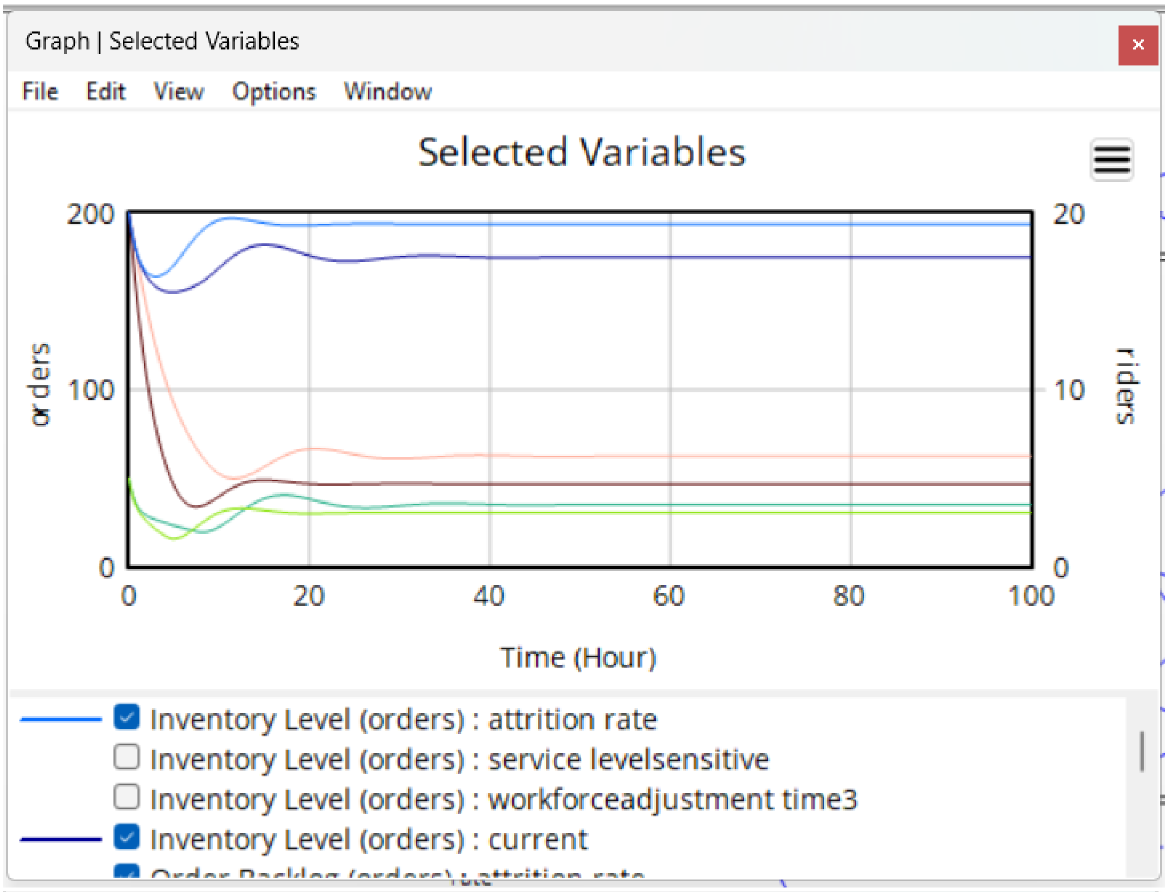
The results show a significant decline in rider workforce, as higher attrition reduces the available delivery capacity. This leads to a sharp increase in order backlog, as the system is unable to fulfill incoming orders efficiently.

Delivery delays increase substantially due to reduced workforce availability, resulting in deterioration of service levels. The system exhibits unstable behavior, with prolonged backlog accumulation and slower recovery.

Inventory levels remain available but cannot be effectively utilized due to limited delivery capacity, indicating a shift in the system bottleneck from inventory to workforce.

Overall, the scenario demonstrates that increased attrition destabilizes the system and significantly impacts service performance.

Real-World Relevance:
This reflects recent situations where delivery partners of platforms like Blinkit and Zepto raised concerns over high work pressure and unsafe delivery expectations. Increased rider exits or protests reduce delivery capacity, leading to delays and backlog accumulation, similar to the behavior observed in the model.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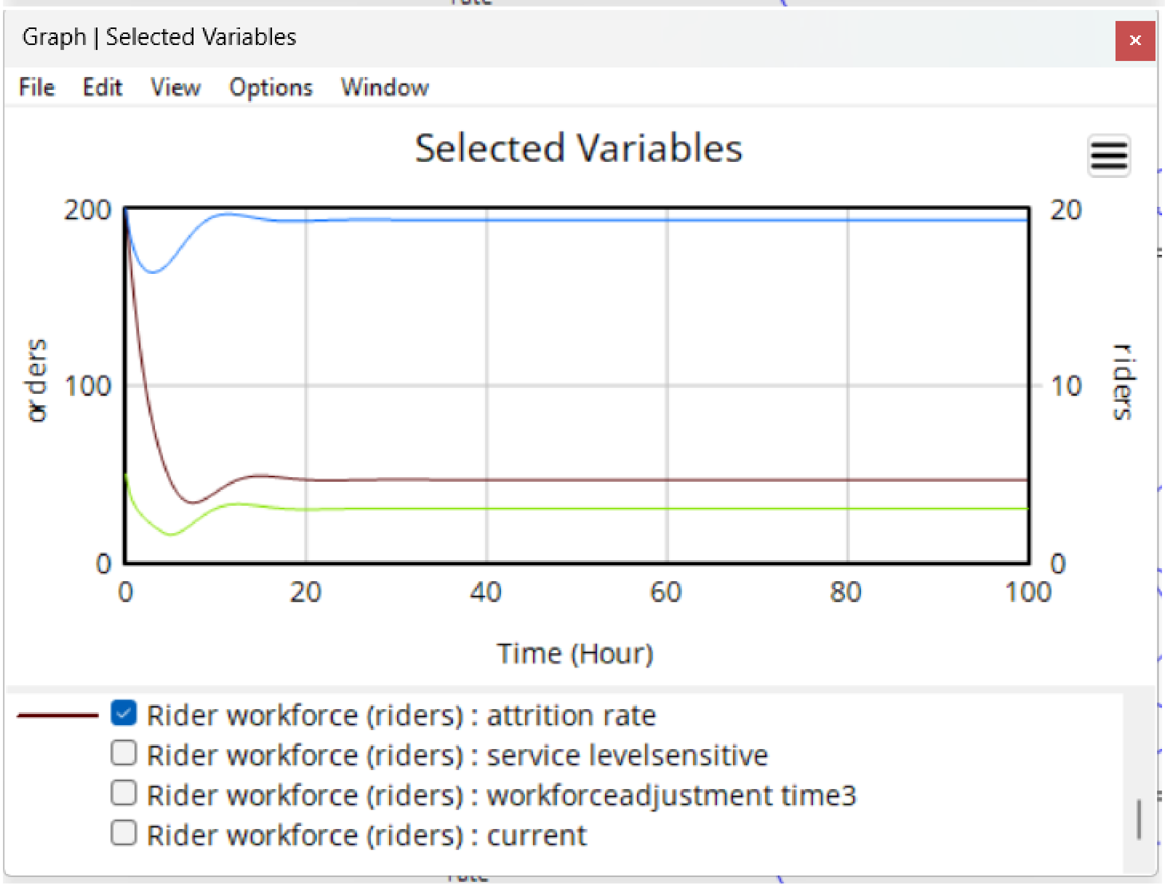
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Rider workforce (riders) : current

What if Scenario 7 : Demand Surge due to Discounts

To analyze the impact of competitive pricing strategies, base demand was increased from 25 to 45 orders per hour to represent demand surges caused by discounts and promotional campaigns, while keeping all other parameters constant.

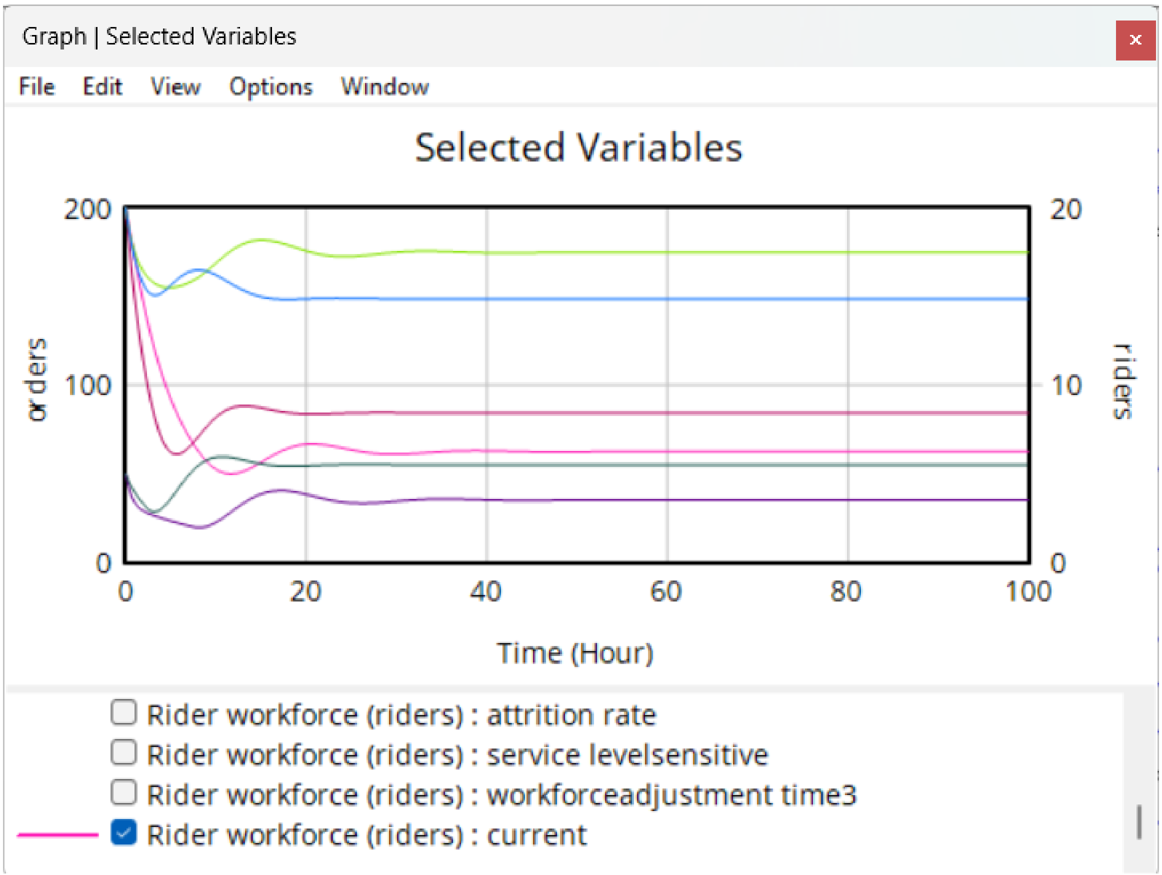
The results show that order backlog increases sharply, with higher peaks and slower stabilization compared to the base case. The system experiences congestion as demand exceeds fulfillment capacity.

Delivery delays rise significantly due to increased capacity utilization, leading to a decline in service levels. The rider workforce increases gradually in response to higher backlog, but due to delays in workforce adjustment, it cannot immediately match the surge in demand.

Inventory levels decline more rapidly due to increased order fulfillment and stabilize at a lower level, indicating higher pressure on replenishment processes.

Overall, the system operates under stressed conditions with reduced performance during demand surges.

Real-World Relevance:
This scenario is commonly observed during festive sales or competitive discount campaigns in platforms like Blinkit and Swiggy Instamart. Sudden increases in order volume lead to delivery delays, rider overload, and temporary service degradation, as companies struggle to scale operations instantly.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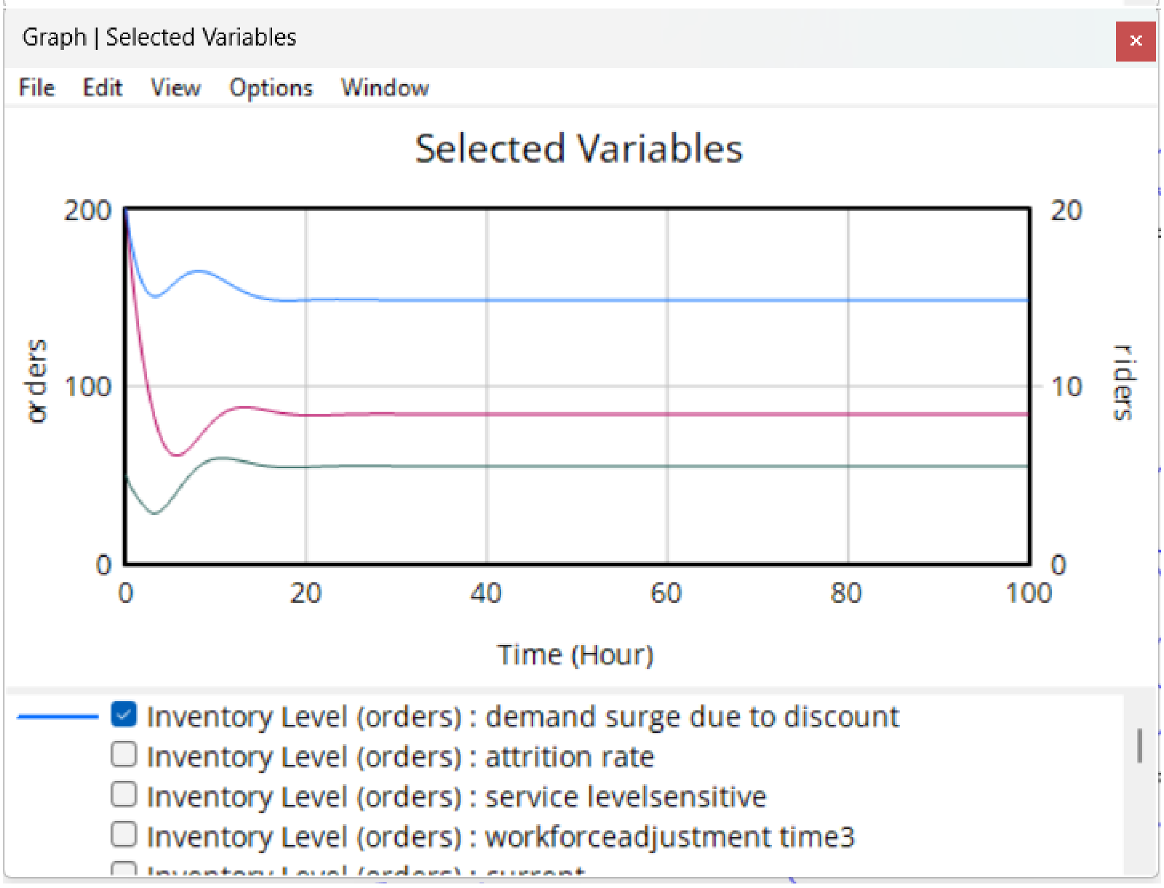
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Time (Hour)
Inventory Level (orders) : demand surge due to discount
Inventory Level (orders) : attrition rate
Inventory Level (orders) : service levelsensitive
Inventory Level (orders) : workforceadjustment time3

Appendix -1

Project Summary

Quick commerce platforms operate under strict 10-20 minute delivery commitments, cre- ating intense pressure on capacity and fulfillment systems. Demand variability, rider availability, picking constraints, and inventory delays interact dynamically, often leading to backlog growth and unstable service levels.

This project develops a System Dynamics model to examine how these feedback effects influence delivery performance over time. The model captures order arrivals, backlog accumulation, capacity scaling, and service-level response within a representative quick commerce network.

Through simulation experiments, the study evaluates alternative capacity and response policies under demand uncertainty. The objective is to identify structural causes of insta- bility and explore strategies that sustain reliable service without excessive overreaction in workforce or inventory decisions.

Detailed Project Description

Quick commerce has grown very rapidly in urban areas, especially with the rise of dark stores that allow deliveries within 10-20 minutes. While this model works well from a customer perspective, it creates operational challenges. Orders fluctuate during the day and spike during peak hours or promotions. At the same time, rider availability, picking capacity inside the dark store, and inventory replenishment cannot adjust immediately.

In such systems, small increases in demand can quickly lead to backlog accumulation. As orders pile up, delivery time increases and customer satisfaction may decline. Managers respond by hiring more riders, extending shifts, or increasing stock levels. However, these adjustments involve delays. Hiring takes time, onboarding takes time, and inventory replenishment depends on supplier lead times. Because of these delays, the system may overreact or underreact, leading to oscillations in backlog, capacity, and service level.

Research in industrial dynamics and supply chain systems shows that delays combined with feedback often generate instability (Forrester, 1961; Sterman, 2000). In last-mile de- livery systems, short delivery windows increase sensitivity to demand fluctuations (Boysen et al., 2021). Urban logistics studies also highlight that operational efficiency depends on how capacity is adjusted over time rather than only increasing resources (Mangiaracina et al., 2019). These findings support the use of System Dynamics to understand structural behavior in quick commerce networks.

For this project, we will build a Stock-and-Flow Diagram (SFD) in Vensim to represent a simplified quick commerce dark store system. The model will focus on three main stocks: order backlog, rider workforce, and inventory level. Backlog increases when order arrival exceeds fulfillment. Rider workforce changes through hiring and attrition. Inventory increases through replenishment and decreases through sales. Order fulfillment will depend on the minimum of available inventory, picking capacity, and rider delivery capacity.

We will include realistic delay structures such as hiring adjustment time and replenish- ment lead time. Delivery delay will be modeled as a function of backlog relative to rider capacity. If backlog becomes too large relative to available riders, delivery delay increases significantly. This delay can influence managerial response intensity, creating feedback between service performance and capacity decisions.

The model is expected to capture three main feedback patterns: a reinforcing demand- pressure loop, a balancing capacity-adjustment loop, and an inventory-control loop. The interaction of these loops, especially under demand surge conditions, may create oscilla- tory behavior similar to classical industrial dynamics models.

We plan to simulate the system over several months and test policies such as faster versus slower hiring responses, different safety inventory levels, gradual capacity scaling, and demand surge scenarios. We will also observe how strict service commitments influence system stability. Performance will be evaluated using average delivery delay, backlog size, rider utilization, inventory stability, and overall responsiveness of the system.

By analyzing simulation results, we aim to identify leverage points that improve stability without excessive cost escalation. The overall goal is to provide a structured understand- ing of how capacity and inventory decisions affect service-level stability in quick commerce delivery networks.

Initial References

1. Forrester, J. W. (1961). *Industrial Dynamics*. MIT Press.
2. Sterman, J. D. (2000). *Business Dynamics: Systems Thinking and Modeling for a Complex World*. McGraw-Hill.
3. Boysen, N., Fedtke, S., & Schwerdfeger, S. (2021). Last-mile delivery concepts: A survey from an operational research perspective. *European Journal of Operational Research*, 289(2), 343-361.
4. Mangiaracina, R., Perego, A., & Tumino, A. (2019). Innovative solutions to increase last-mile delivery efficiency in urban areas. *Sustainability*, 11(9).

Appendix - 2

1. Project Summary

Quick commerce platforms operate under strict **10–20 minute delivery commitments**, creating significant pressure on logistics capacity and fulfillment operations. Demand variability, rider availability, picking constraints, and inventory replenishment delays interact dynamically, often leading to backlog accumulation and unstable service levels.

This project develops a **System Dynamics (SD) model** to examine how these feedback effects influence delivery performance over time. The model represents a simplified quick-commerce dark store system where order arrivals, backlog accumulation, workforce adjustments, and inventory dynamics interact through feedback loops.

Using simulation experiments, the study evaluates how operational policies such as workforce adjustment speed, inventory replenishment delays, and demand fluctuations influence service performance. The goal is to identify **structural causes of instability** and explore strategies that maintain reliable service levels without excessive workforce expansion or inventory overstocking.

2. System Description

Quick commerce has grown rapidly in urban environments, particularly with the emergence of **dark stores**, which are small warehouses dedicated to fast fulfillment of online orders.

Although customers benefit from rapid delivery promises, the operational system faces several challenges:

- Demand fluctuates throughout the day and spikes during peak periods.
- Rider workforce cannot adjust instantly.
- Inventory replenishment involves supplier lead times.
- Order picking and dispatch processes introduce delays.

When order arrivals exceed fulfillment capacity, **order backlog increases**, which leads to longer delivery delays and declining service levels. Managers often respond by hiring additional riders or increasing inventory levels. However, these responses occur with delays, which may lead to oscillations in backlog and service performance.

System Dynamics provides an effective approach to analyze such systems because it captures the **feedback loops and time delays** that govern system behavior.

3. Model Assumptions

To simplify the complex operations of quick commerce delivery networks, several assumptions were made while developing the system dynamics model.

Aggregate System Representation

The model represents the quick commerce system at an **aggregate level** rather than modeling individual orders, riders, or products. All orders and deliveries are treated as average flows over time.

Constant Rider Productivity

Each rider is assumed to deliver a fixed average number of orders per hour. Productivity does not vary due to traffic conditions, weather, or rider experience.

Homogeneous Orders

All customer orders are assumed to require similar picking time and delivery effort. Differences in order size, product types, or delivery distance are not explicitly modeled.

Continuous Demand

Customer demand is modeled as a continuous flow of orders per hour rather than discrete individual orders.

Hiring and Attrition Dynamics

Workforce adjustments occur gradually based on the workforce adjustment time. Riders leave the system according to a fixed attrition fraction.

Inventory Replenishment Policy

Inventory replenishment is based on the gap between target inventory and current inventory level. Replenishment orders are assumed to arrive after a fixed lead time.

Fulfillment Constraints

Order fulfillment is constrained by three factors: available backlog, available inventory, and delivery capacity. The system fulfills orders based on the minimum of these constraints.

Delivery Delay Representation

Delivery delay is assumed to increase as capacity utilization rises. Higher utilization indicates operational congestion, which leads to longer delivery times.

Fixed Parameter Values

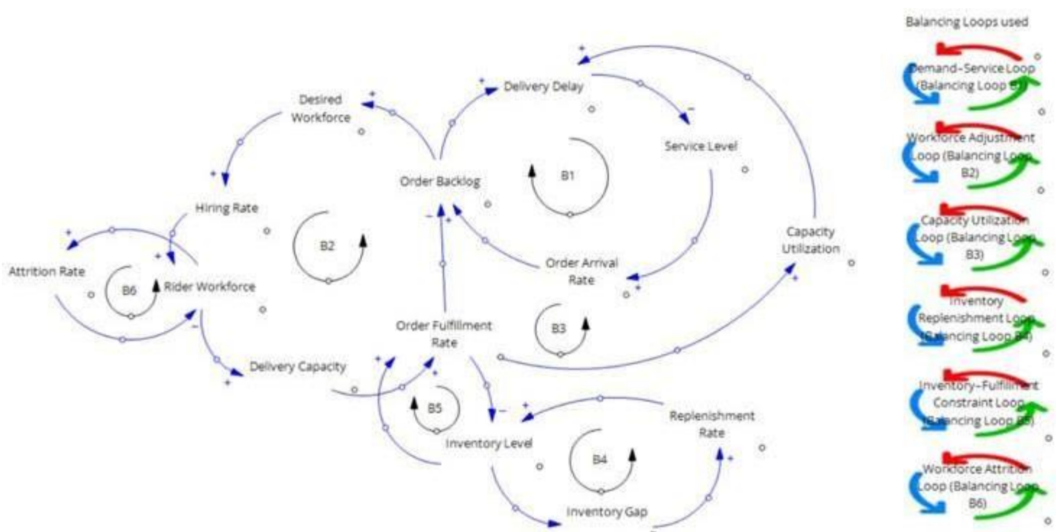
Key parameters such as base demand, target delay, inventory lead time, and workforce adjustment time are assumed to remain constant during the simulation period.

No External Disruptions

The model does not explicitly consider external disruptions such as traffic congestion, system failures, weather conditions, or sudden supply chain disruptions.

4. Causal Loop Diagram (CLD)

To understand the feedback structure of the system, a **Causal Loop Diagram (CLD)** was developed.



The CLD highlights several interacting **balancing feedback loops** that regulate system performance.

The causal loop diagram illustrates the feedback mechanisms governing demand, workforce capacity, delivery performance, and inventory dynamics in the quick commerce delivery network.

B1: Demand–Service Stabilization Loop

This loop represents the interaction between delivery performance and customer demand. When the order backlog increases, delivery delays rise due to higher operational pressure on the system. Longer delivery delays reduce service quality, which may discourage future order arrivals. As demand decreases, the backlog gradually stabilizes. This feedback mechanism helps regulate system demand when service performance deteriorates.

B2: Workforce Adjustment Loop

The workforce adjustment loop captures managerial responses to increasing backlog levels. When the backlog grows, the system estimates a higher workforce requirement and increases the hiring rate. As more riders are added, delivery capacity expands and order fulfillment improves. This increased capacity reduces the backlog and helps restore system balance.

B3: Capacity Utilization Loop

This loop represents operational congestion in the delivery network. When order fulfillment increases relative to available delivery capacity, capacity utilization rises. High utilization increases delivery delays due to operational strain on riders and logistics resources. Increased delays reduce service performance, which influences demand and eventually stabilizes system workload.

B4: Inventory Replenishment Loop

Inventory levels are maintained through a replenishment policy based on the gap between target inventory and actual inventory levels. When inventory falls below the desired level, replenishment orders are triggered to restore stock levels. Over time, this process stabilizes inventory availability in the dark store.

B5: Inventory–Fulfillment Constraint Loop

Inventory availability directly affects the system’s ability to fulfill orders. As orders are dispatched, inventory levels decrease. If inventory becomes insufficient, it constrains the fulfillment rate and slows down order processing. This feedback ensures that fulfillment activity remains consistent with available stock levels.

B6: Workforce Attrition Loop

The workforce system also includes natural attrition dynamics. As the number of riders increases, attrition also rises due to turnover or operational fatigue. This reduces the rider workforce over time and requires continuous hiring to maintain adequate delivery capacity.

Empirical Basis for Feedback Structure

The feedback loops incorporated in the quick commerce delivery model are motivated by observed operational patterns in ultra-fast delivery platforms. Industry reports and recent market developments indicate that service reliability, workforce flexibility, delivery density, and inventory responsiveness are key determinants of platform stability. These real-world insights are mapped to specific feedback loops in the model as shown below.

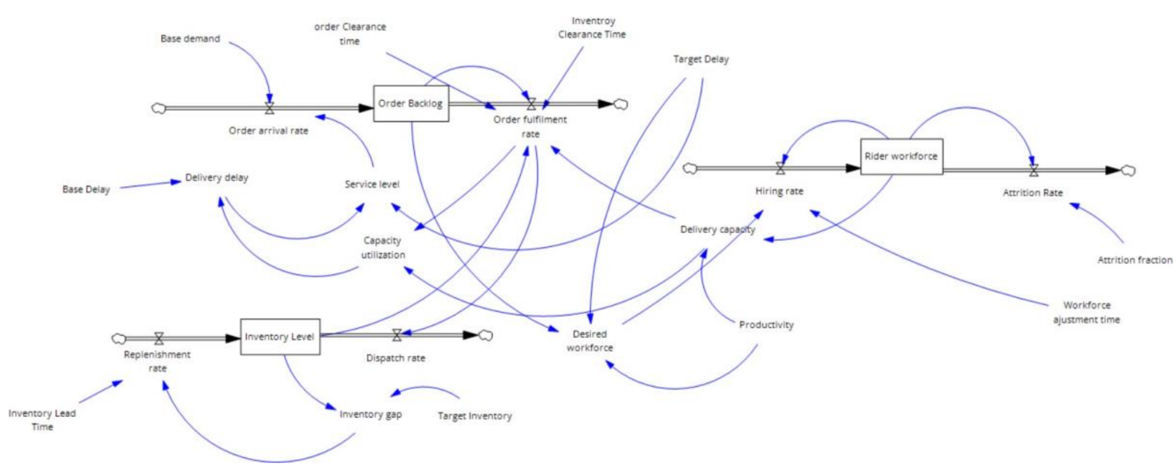
Mapping of Model Feedback Loops to Industry Evidence

Feedback Loop in Model	Behaviour Captured	Real Industry Observation	Reference
B1: Demand–Service Stabilization Loop	Rising backlog increases delivery delay and reduces service level, which moderates future order arrivals	Customer demand in rapid delivery platforms is highly sensitive to delivery reliability and waiting time consistency	Roland Berger (2021) Quick Commerce Report
B2: Workforce Adjustment Loop	Increasing backlog leads to higher workforce requirements and gradual hiring, expanding delivery capacity over time	Quick commerce firms experience labour scaling challenges and onboarding delays during demand surges	Roland Berger (2021)
B3: Capacity Utilization Loop	Higher fulfillment relative to available capacity increases operational congestion, leading to longer delivery delays	Rapid expansion phases have resulted in rider overload and unstable delivery performance in dark store networks	Gorillas Strategy Case Study

B4: Inventory Replenishment Loop	Inventory shortages trigger replenishment actions that restore stock levels and stabilize fulfillment operations	High-frequency demand environments require responsive restocking policies and safety inventory buffers	Khurana and Jain (2025) Quick Commerce Inventory Study
B5: Inventory–Fulfillment Constraint Loop	Limited inventory availability directly restricts the order fulfillment rate and slows backlog clearance	Stockouts in ultra-fast delivery platforms reduce service reliability and increase customer dissatisfaction	Industry operational observations summarized in consulting reports
B6: Workforce Attrition Loop	Increasing workforce size also increases attrition pressure, requiring continuous hiring to maintain delivery capacity	Rider turnover and retention challenges are commonly observed in gig-based delivery operations	Industry labour trend analyses

5. Stock-and-Flow Diagram (SFD)

The CLD relationships were translated into a **Stock-and-Flow Diagram (SFD)** using Vensim.



The SFD captures the quantitative structure of the system.

6 System Dynamics Model Formulation Time

Unit model - Hour

1. Stock Variables

Order Backlog (Unit: orders)

Order Backlog = INTEG (Order Arrival Rate - Order Fulfilment Rate , 50)

Rider Workforce (Unit: riders)

Rider Workforce = INTEG (Hiring Rate - Attrition Rate , 20)

Inventory Level (Unit: orders)

Inventory Level = INTEG (Replenishment Rate - Dispatch Rate , 200)

2. Flow Variables

Order Arrival Rate (Unit: orders/hour)

Order Arrival Rate = Base Demand * Service Level

Order Fulfilment Rate (Unit: orders/hour)

Order Fulfilment Rate = MIN (Order Backlog / Order Clearance Time ,Inventory Level / Inventory Clearance Time ,Delivery Capacity)

Hiring Rate (Unit: riders/hour)

Hiring Rate = (Desired Workforce - Rider Workforce) / Workforce Adjustment Time

Attrition Rate (Unit: riders/hour)

Attrition Rate = Rider Workforce * Attrition Fraction

Replenishment Rate (Unit: orders/hour)

Replenishment Rate = Inventory Gap / Inventory Lead Time

Dispatch Rate (Unit: orders/hour)

Dispatch Rate = Order Fulfilment Rate

3. **Auxiliary Variables**

Delivery Capacity (Unit: orders/hour)

Delivery Capacity = Rider Workforce * Productivity

Productivity (Unit: orders/(hour*riders)) Productivity = 3

Capacity Utilization (Unit: dimensionless)

Capacity Utilization = Order Fulfilment Rate / Delivery Capacity

Delivery Delay (Unit: hours)

Delivery Delay = Base Delay * (1 + Capacity Utilization * Capacity Utilization)

Service Level (Unit: dimensionless) Service Level

= Target Delay / Delivery Delay **Desired**

Workforce (Unit: riders)

Desired Workforce = Order Backlog / (Target Delay * Productivity)

Inventory Gap (Unit: orders)

Inventory Gap = Target Inventory - Inventory Level

4. PARAMETERS

Base Demand = 25 (orders/hour) Base
Delay = 1 (hours)
Target Delay = 1.5 (hours)
Workforce Adjustment Time = 5 (hours) Attrition
Fraction = 0.05 (1/hour)
Target Inventory = 250 (orders)
Inventory Lead Time = 4 (hours)

Inventory Clearance Time = 1 (hours)

7. Simulation Discussion

The system dynamics model reveals several important operational behaviors.

Backlog Growth During Demand Surges

When demand suddenly increases, order arrival rate exceeds fulfillment capacity. As a result:

- backlog grows rapidly
- delivery delay increases
- service level deteriorates

Because workforce expansion requires time, the system cannot immediately respond to demand shocks.

Workforce Adjustment Delays

Hiring delays create oscillatory dynamics in the system. If

workforce adjustment time is long:

- backlog remains high for extended periods
- delivery delay fluctuates
- service performance becomes unstable

This behavior is consistent with classical system dynamics models of workforce adjustment.

Inventory Constraints

Inventory availability also limits order fulfillment. If

inventory replenishment lead times are large:

- stockouts occur
- fulfillment rate drops
- backlog increases even if riders are available

This highlights the importance of **inventory policies in quick commerce systems**. **Capacity**

Utilization Effects

High capacity utilization leads to operational congestion. When

riders operate close to maximum capacity:

- delivery delays increase
- service quality decreases
- demand may decline temporarily

This feedback mechanism helps stabilize the system but also creates temporary service disruptions.

8. Policy Insights

The model provides insights into how quick commerce systems can maintain stable service levels.

Faster Workforce Adjustment

Reducing workforce adjustment time allows faster hiring, reducing backlog spikes.

Inventory Safety Stock

Maintaining higher safety inventory reduces stockouts and stabilizes fulfillment.

Gradual Capacity Expansion

Instead of aggressive hiring, gradual workforce scaling helps avoid oscillations.

9. Conclusion

This study developed a System Dynamics model to analyze capacity and service-level stability in quick commerce delivery networks. The model captures the interactions between order demand, workforce dynamics, delivery capacity, and inventory management.

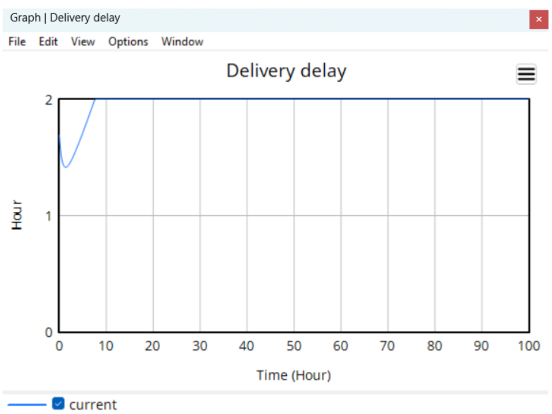
The analysis shows that system instability often arises from **delays in workforce adjustment and inventory replenishment**. These delays interact with demand fluctuations to produce backlog growth and service degradation.

By understanding the feedback structures governing the system, managers can design policies that improve stability while avoiding excessive operational costs.

The simulation results show that the system initially experiences fluctuations due to adjustment delays in workforce and inventory replenishment. Over time, the interacting feedback loops stabilize the system, leading to equilibrium levels of backlog, delivery delay, workforce size, and inventory availability.

10. Graphs and Interpretations

The order backlog initially fluctuates as the system adjusts from its initial conditions. As rider workforce and delivery capacity adapt to demand, the backlog gradually stabilizes at a steady level, indicating that fulfillment capacity has balanced incoming orders.



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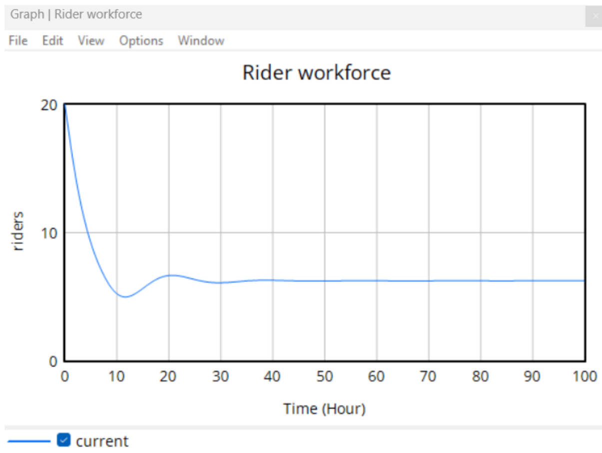
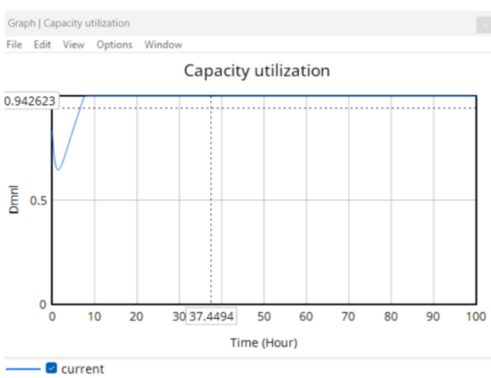


Rider workforce initially declines due to attrition and adjustment dynamics before stabilizing near the required capacity level. This reflects the delayed workforce adjustment mechanism responding to operational demand.

Inventory levels initially decline due to order fulfillment but gradually recover as replenishment responds to the inventory gap. Over time, the system stabilizes near the target inventory level.



Capacity utilization rises quickly as order demand approaches system capacity. As workforce and fulfillment capacity adjust, utilization stabilizes close to its equilibrium level, indicating efficient use of delivery resources.

11. Initial References

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